

HydroSense-Kenya

An Integrated Scientific Computing Platform for Subsurface Hydrology
and Automated Irrigation Management

Author: Trevor Cecil

Course: BSc. Computer Science

Institution: Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Instructor / Supervisor: Dr. Nderu

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Executive Summary

HydroSense-Kenya addresses critical challenges in precision agriculture by uniting modern data engineering practices with classical numerical analysis methods. Developed as a fully decoupled, modular system, the software architecture systematically consumes noisy, unreliable environmental and soil agricultural IoT data streams, isolates hardware malfunctions, derives survival thresholds via numerical root-finding algorithms, and simulates subsurface soil moisture kinetics using high-order differential equations. The system culminates in an integrated management control suite designed to automate resource allocation efficiently across distinct farming zones.

1. Software Architecture and Reproducibility

To meet strict production deployment and programmatic evaluation standards, the codebase follows an explicit Separation of Concerns design pattern. Presentation layers, experimental scripts, and diagnostic exploratory sandboxes are strictly isolated into independent runtime environments.

```
HydroSense-Kenya/  
  ■■■ data/  
    ■ ■■■ raw/           # Unprocessed IoT telemetry data files  
    ■ ■■■ processed/     # Sanitized, merged datasets (CSV standard)  
    ■■■ src/             # Pure algorithmic backend engines  
    ■ ■■■ data_cleaning.py # Ingestion, structural dictionary & cleaning  
    pipelines  
    ■ ■■■ numerical_methods.py # Root-finding and differential ODE simulation  
    engines  
    ■■■ notebooks/       # Structural verification and integration  
    interfaces  
    ■■■ tests/           # Regression and unit testing matrix via Pytest
```

2. Telemetry Ingestion and Data Sanitation

Raw telemetry is vulnerable to missing data blocks, literal value transmission faults, and physical clipping spikes. The ingestion pipeline executes a multi-tiered sanitation pattern prior to analytical evaluation:

Level 1 Ingestion Mapping Rules

The pipeline programmatically parses unassigned text spaces and literal string values such as 'NA' into mathematical null representations (`numpy.nan`). This allows down-stream numerical vectors to track missing dimensions safely without raising systematic type exceptions.

Time-Series Realignment and Outlier Remediation

Physical sensor limits are enforced to purge environmental anomalies. Thermal records exceeding 40°C (such as an impossible 45.8°C transmission failure) and water storage numbers scaling over 5000 liters (such as a 9900L capacity overflow) are flagged as invalid. Crucially, multi-zone arrays are explicitly sorted chronologically by timestamp prior to executing linear time-series interpolations. This prevents cross-zone timeline bleeding and protects chronological integrity.

3. Algorithmic Root-Finding Core

Rather than relying on static bounds, HydroSense-Kenya establishes dynamic critical operational thresholds. The mathematical baseline solves for the exact temperature boundaries where crop evapotranspiration outpaces metabolic thresholds under current ambient conditions:

$$f(T) = 0.12T + 0.35W + 2.4S - 0.025H - ET_{critical} = 0$$

Implemented Numerical Models:

Bisection Method: Robust bracket root finding constrained over an initial intervals of [10, 50], providing guaranteed convergence stability.

Newton-Raphson Method: High-speed quadratic tracking utilising an analytically defined derivative vector of $f'(T) = 0.12$.

Secant Method: Superlinear convergence tracking that approximates scaling slope transitions without requiring direct structural derivatives.

4. Dynamic Subsurface Simulation

To trace subsurface hydrology patterns, the system treats soil structures as a continuous differential matrix. Volumetric variations are modeled dynamically based on water influx against ambient crop extractions:

$$dS/dt = \text{Rainfall}(t) + \text{Irrigation}(t) - \text{Drainage}(S, t) - \text{Evapotranspiration}(T, t)$$

To solve this continuous relationship over iterative discrete bounds, two execution strategies were developed and implemented: Euler's First-Order Approximation for foundational state processing, and a Runge-Kutta 4th Order (RK4) engine to mitigate local truncation errors and maintain system stability during highly fluctuating meteorological periods.

5. System Verification and Compliance

System reliability is systematically evaluated via an automated test suite containing 9 independent test blocks. Core computational engines are verified under rigid mathematical boundaries:

Test Engine Module	Target Mathematical Focus	Tolerance Matrix	Status
test_root_finding.py	Bisection / Newton-Raphson / Secant	atol = 1e-4	PASSED
test_integration.py	Trapezoidal / Simpson Inflow Rules	atol = 1e-4	PASSED
test_linear_systems.py	Gaussian Network Flow Reductions	atol = 1e-5	PASSED
test_simulation.py	Euler / Runge-Kutta 4 Integrations	atol = 1e-4	PASSED

Conclusion

HydroSense-Kenya proves that decoupling complex programmatic operations into strict library scripts, validated by robust unit testing matrixes, results in an elite, adaptable scientific framework. The dynamic threshold modeling pipeline successfully matches core environmental ground truths, establishing a secure architecture for smart-infrastructure agricultural management.